St. John Ambulance

SAVING LIVES
at work, home and play

FIRST AID
REFERENCE GUIDE







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Bell Let's Talk

St. John Ambulance Canada would like to thank Bell Let's Talk for its funding and support of our mental health awareness project.

By including mental health and wellness material in this guide, we are opening the conversation about mental health in first aid classes across Canada.

St. John Ambulance
SAVING LIVES
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FIRST AID

St. John Ambulance Saint-Jean

SAVING LIVES at work, home and play

SAUVER DES VIES au travail, à la maison et dans les loisirs

REFERENCE GUIDE

Fourth Edition

St John
St-Jean

Fourth edition – January 2019

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Medical advisory group for St. John Ambulance

St. John Ambulance is a leading provider of first aid training in Canada offering its courses to over half a million Canadians each year. St. John Ambulance is committed to providing Canadians with first aid training that is based on scientific evidence and that reflects emerging trends in first aid and emergency care. A Medical Advisory Group has been established to support the adoption of evidence based practice and to contribute to shaping the future of first aid by providing technical expert advice to St. John Ambulance.

The purpose of the Medical Advisory Group is to a) provide advice and recommendations to SJA on the appropriateness of first aid practices in accordance with the scientific evidence and b) advise on emerging trends in first aid and emergency care that may impact SJA programs.

Below are the names and credentials of the members of our Medical Advisory Group. We are proud to have them as part of our community.

- Dr. Robert Boyko, MD, CCFP(EM), FCFP
- Dr. Trevor Jain, BSc, MD, CCFP(EM), MSc
- Dr. Brian Metcalfe, BSc, MD, CCFP(EM)
- Dr. Jonathan Wallace, PCP, BSc, MD, CCFP(EM,FPA) FRACGP
- Steven Anas, PCP
- Kevin Morgan, BAHSc, A-EMCA

St. John Ambulance

St. John Ambulance is a charity and international humanitarian organization dedicated to helping Canadians improve their health, safety and quality of life through training and community service. Revenue generated from first aid/CPR training supports St. John Ambulance's charitable work in Canada and around the world.

Coast-to-coast, more than 12,000 front-line volunteers serve communities by providing first aid services at public events and during emergencies. St. John Ambulance volunteers also improve quality of life through programs that help seniors, the disadvantaged, and youth.

As Canada's leading authority in first aid and CPR services since 1883, St. John Ambulance offers innovative programs and products to save lives at work, home, and play.

To contact your local St. John Ambulance, visit www.sja.ca.

Fast facts

- Established in 1883 in Canada with roots going back 900 years
- Issues over 500,000 certificates in first aid and CPR to Canadians each year
- Supports humanitarian relief efforts across Canada and around the world
- St. John Ambulance front-line volunteers provide service at approximately 10,000 public events annually
- More than 1 million volunteer hours of community service are provided annually
- More than 100,000 individuals were assisted by St. John Ambulance volunteers in 2014
- 365 Canadians were formally recognized for their efforts to save a life in 2015
- Approximately 3,000 St. John Therapy Dog teams provided over 200,000 volunteer hours in 2014

This Reference Guide was developed in accordance with the 2016 Canadian Consensus Guidelines on First Aid and CPR, an evidence-based set of recommendations on training and standards of practice for first aid and CPR. The guidelines are released by the Canadian Guidelines Consensus Task Force comprised of the Heart and Stroke Foundation, the Canadian Red Cross, St. John Ambulance, the Canadian Ski Patrol, and the Lifesaving Society.

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The information ("Information") provided in this book is for general use and knowledge and does not contain all information that may be relevant to every situation. The Information cannot be relied upon as a substitute for seeking guidance from appropriate professionals, such as physicians. Users acknowledge and agree that St. John Ambulance is not responsible or liable for the user's actions or decisions resulting from the information (including information regarding medication or other drugs) in this book, including but not limited to choosing to seek or not to seek advice from medical professionals such as physicians.

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Emergency Phone Numbers

Police	_____
Fire	_____
Ambulance	_____
Poison Control	_____
Emergency Contact	
Name	_____
Phone	_____
Home Phone	_____
Street Address	_____

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Chapter 1

Introduction to First Aid



- Roles and responsibilities
- First aid n the workplace
- First aid and the law
- Safety and personal protection
- Signs and symptoms and mechanism of injury
- The importance of medical help
- Stress management in emergency situations

1

Chapter 1 Introduction to First Aid

This guide covers a wide range of information that will help you respond appropriately in a first aid or medical emergency. The introductory chapter contains background information, definitions and other material related to giving first aid. Chapter 2 explains casualty management including issues that relate to assessment of the casualty. This chapter also includes topics that are important to understand in the first critical moments at the emergency scene. Chapter 5 deals with issues of particular interest to health care providers—responders with a specific duty to respond within the health care system.

This guide is used to support the teaching of these and other courses:

- Standard and Emergency First Aid
- Basic and Intermediate First Aid
- CPR at all levels including Basic Life Support/Health Care Provider (BLS/HCP)

Some content contained in this guide will not pertain specifically to the level of training you have received. First aiders should always remember not to exceed their training, or the regulations/legislation of their province or territory.

Roles and responsibilities

What is first aid?

First aid is emergency help given to an injured or suddenly ill person using readily available materials. A person who takes charge of an emergency scene and gives first aid is called a first aider. The injured or ill person is called a casualty.

The three priorities of first aid, in order of importance, are to:

- Preserve life
- Prevent the illness or injury from becoming worse
- Promote recovery

Introduction to First Aid

What can a first aider do?

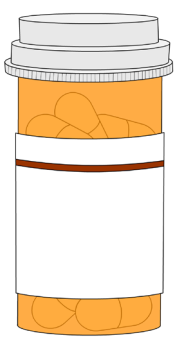
First aiders do not *diagnose or treat injuries or illnesses* (except, perhaps, when they are very minor)—this is what medical doctors do. A first aider *suspects injuries and illnesses*, and *gives first aid* at the scene.

Besides giving first aid, it is important to:

- Protect the casualty's belongings
- Keep unnecessary people away
- Reassure family or friends of the casualty
- Clean up the emergency scene and work to correct any unsafe conditions that may have caused the injuries in the first place

First aiders within a workplace may have obligations and protections under federal, provincial, and territorial legislation regarding administering medications. Refer to federal, provincial, or territorial legislation and regulations for the requirements in your area.

When assisting with medications, the first aider should check the label and ensure the following "5-rights" are met:



- Right **Person** – the name of the casualty is the name on the medication
- Right **Medication** – is this the right medication for this situation?
- Right **Amount** – what are the dosing instructions
- Right **Method** – how is this medication to be taken? (By mouth, inhaled, etc.)
- Right **Time** – is this the right time to take this medication?

1

A casualty's age in first aid and CPR

The procedures related to the provision of first aid and cardiopulmonary resuscitation (CPR) differ in some ways, depending on the age and size of the casualty.

In first aid and CPR:

- An **infant** casualty is under one year old
- A **child** casualty is from age one to age eight
- An **adult** casualty is over eight years of age

It is important to recognize that these ages are guidelines. The size of the casualty must be considered.

First Aid in the Workplace

Workplace first aid is emergency care given by a person who is trained (and designated) to be the first aid provider to a co-worker who is injured or suddenly ill. All provinces and territories have a provision within legislation relating to first aid in the workplace. Refer to your provincial or territorial regulations for what is required in your area.

Giving first aid as part of your job

When giving first aid is part of your job, you have a duty to use reasonable skill and care based on your level of training. There are regulations to protect the first aider. For example in workplaces regulated by the Canada Labour Code, Section 126(3) of the Code states:

"No employee is personally liable for anything done or omitted to be done in good faith by the employee when the employee is assisting the employer, as requested by the employer, in providing first-aid or in carrying out any other emergency measures."

Every workplace in Canada is required under federal, provincial, or territorial regulations to have a first aid kit. The size and contents of the first aid kit will be determined by those regulations. First aid providers are encouraged to be familiar with the contents of their workplace first aid kit, its location, and to conduct regular inspections of the kit.

First Aid as part of OHS

Provincial and territorial regulations and legislation contain requirements for first aid training in their jurisdiction. Most regulations require at least one designated first aid provider at work at all times. The level of training required by the first aid provider will depend on the size of the workplace, the distance to medical help, and the risk of injury in the workplace.

To help ensure compliance with regulations, it is recommended to have more than one person on each shift trained in first aid to account for holidays, illness, and breaks.

OHS Legislation

Provinces and territories are responsible for establishing Occupational Health and Safety legislation to protect workers. All workplaces that fall under provincial or territorial jurisdiction concerning regulations are required to adhere to the legislation and regulations of that province or territory. Those work places that fall under federal jurisdiction are subject to the Canada Labour Code.

Housed within the legislation and regulations are provisions for adequate first aid coverage for a workplace, usually based on some or all of the following:

- The number of workers
- The potential risks
- The distance from medical care

First aid and the law

Note that St. John Ambulance is not giving legal advice. This guide is not intended to replace advice given by a lawyer or legal professional.

Principles of the Good Samaritan

Across Canada Good Samaritan laws and principles protect first aiders from lawsuits. You are a Good Samaritan if you are a bystander who helps a person when you have no legal duty to do so. As a Good Samaritan, you give your help without being paid, and you give it in good faith. Whenever you help a person in an emergency situation, you should abide by the following principles:

- You identify yourself as a first aider and get permission to help the injured or ill person before you touch them—this is called **consent**
- You use **reasonable skill and care** in accordance with the level of knowledge and skill that you have
- You are not **negligent** in what you do
- You do not **abandon** the person

Consent

The law says everyone has the right not to be touched by others. As a first aider, you must respect this right.

Always ask if you can help. If the casualty cannot answer, you have what is called **implied consent,** and you can help.

If the casualty is an infant or a young child, you must get consent from the child's parent or guardian. If there is no parent or guardian at the scene, the law assumes the casualty would give consent if they could, so you have implied consent to help.

A person has the right to refuse your offer of help. In this case, do not force first aid on a conscious casualty. If you do not have consent to help, there may be other actions you can take without touching the casualty, such as controlling the scene, and calling for medical help.

Be aware of difficulties in communicating when a casualty:

- Is hard of hearing
- Speaks a different language
- Is visually impaired
- Is a child
- Is in pain
- Shows signs of mood disorder

Reasonable skill and care

As a Good Samaritan, when you give first aid you are expected to use reasonable skill and care according to your level of knowledge and skills.

Negligence

Give only the care that you have been trained to provide, and always act in the best interest of the casualty.

Abandonment

Never abandon a casualty in your care. Stay until:

- You hand them over to medical help
- You hand them over to another first aider
- They no longer want your help—this is usually because the problem is no longer an emergency, and further care is not needed



Check your applicable workplace legislation/ regulations

Giving first aid in Quebec

The Quebec Charter of Human Rights and Freedoms declares that any person whose life is in danger has the right to be helped. This means that you are required to help a person whose life is at risk, provided you do not put your own life, or anyone else's, in danger.

Safety and personal protection

In any emergency, first aid providers must always be aware of hazards and give first aid safely. A hazard is anything that poses a risk of injury or death to a first aid provider. There are three basic types of risks to be aware of:

- The energy source that caused the original injury—is the mechanism that caused the original injury still active, causing injury to others? Example: where an injury has been caused by machinery, is the machinery still running?
- There may be hazards caused by external factors. Example: passing vehicles may pose a risk at the scene of a motor vehicle incident
- There may be hazards associated with first aid procedures or a rescue. Example: moving a heavy casualty could place the first aider at risk of injury

Some hazards can be controlled by the first aid provider. When controlling hazards, keep the following principles in mind:

- Use mechanical means whenever possible (broom, dustpan, tools, etc.)
- Be careful when lifting or moving objects on or near a casualty
- Have someone assist you where possible
- Turn on lights where no other risks to doing so exist
- Ensure safe footing – many injuries to first aiders are a result of slips and falls.

Hazards that require specialized training to control (electrical hazards, fire, gases, etc.) should only be controlled by those who are properly trained. Most workplaces that deal regularly with these types of hazards will have a specialized response team. Refer to your workplace's policies and protocols.

When dealing with chemical hazards, a Safety Data Sheet (SDS/MSDS) should be accessible and will provide information on how to control the hazards along with first aid directions.

Preventing infection

Airborne pathogens

Examples of infections that can be spread through the air are:

- **Meningitis** is a bacterial or viral infection which causes swelling that affects the spinal cord and brain
- **Tuberculosis** is a bacterial infection that primarily affects the lungs, but can affect any part of the body
- **Influenza**, or "the flu," is a viral infection which is easily spread, and can vary from being mildly debilitating to fatal

Body fluid and blood-borne pathogens

Exposure to blood or body fluids (i.e. vomit, feces) poses a health risk to first aiders. There are three diseases that first aiders should be aware of:

- **Human immunodeficiency virus (HIV)** is the virus responsible for AIDS. There is no vaccine to protect people from this virus. The best defence remains adequate

protection to help prevent infection.

- **Hepatitis B** is one of the three common forms of hepatitis, a viral disease that can cause severe liver damage. Some people who have Hepatitis B have no symptoms but are still contagious. There is a vaccine to prevent Hepatitis B.
- **Hepatitis C** causes much of the same liver damage as Hepatitis B, but there is currently no vaccine available to prevent this disease.

Sharp objects

If a sharp object touches infected blood and then pricks or cuts your skin, you could become infected. First aiders do not use sharp objects like scalpels and needles, but there may be broken glass or other sharp objects that have been in contact with blood or other bodily fluids. Always wear gloves and handle sharp objects with extreme care.

Personal Protective Equipment

Personal Protective Equipment (PPE) is clothing and equipment used to protect the first aider and to minimize the risks of health and safety hazards when in contact with a casualty. PPE can be gloves, a pocket mask used for ventilations, a helmet, eye protection, safety boots, etc.

Use a face mask or shield when providing artificial respiration or CPR. Always follow the manufacturer's directions for disinfecting and cleaning reusable items. Single-use masks, one-way valves, and gloves are disposed of by double bagging with other contaminated articles. If used in the workplace, follow provincial/territorial and/or company protocols for disposal of hazardous items.

Disposable gloves prevent direct hand contact between the first aider and the casualty. Wear gloves when you might touch blood, bodily fluids, tissue or anything that has come in contact with one of these.

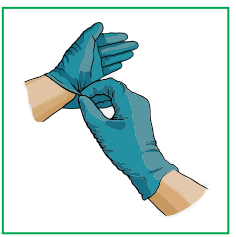


If you tear a glove, wash your hands as soon as possible, and put on a new pair. Dispose of contaminated gloves by sealing them in a plastic bag and double-bagging them.

1

How to remove gloves

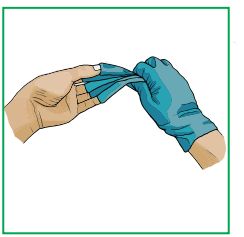
Once gloves have been used, they are contaminated and are a possible source of infection. Take them off without touching their outer surface following the steps below.



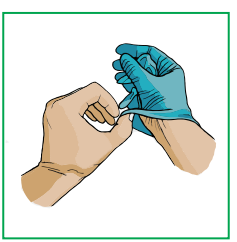
Grasp the cuff of one glove



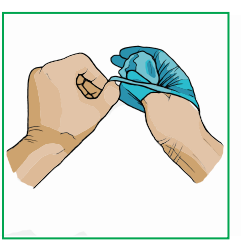
Pull the cuff towards the fingers, turning the glove inside out.



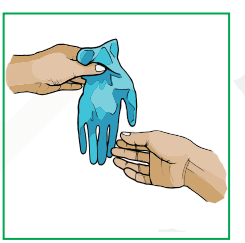
As the glove comes off, hold it in the palm of your other hand.



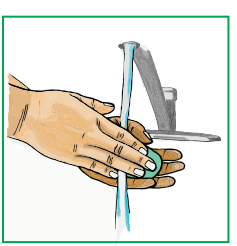
Slide your fingers under the cuff of the other glove.



Pull the cuff towards the fingers over the first glove.



Tie a knot in the top of the outer glove and dispose of properly—see below.



Wash your hands with soap and running water as soon as possible.

Proper Disposal

Seal the used gloves in a plastic bag and put them in your household garbage.

Check with health professionals or your first aid instructor for specific regulations in your area.

Protecting the first aider

Areas of the body that may have come into contact with a casualty need to be cleaned with hot, soapy water, an anti-septic solution, or a mixture of bleach and water (at a ratio of 1:10). Spills should be cleaned, then sanitized with the bleach and water solution for 20 minutes.

Anyone who has been exposed to possible contaminants should take a hot shower with soap and rinse thoroughly. Anyone who has been exposed to contaminants from a needle stick or sharps injury should seek medical attention.

If an injury occurs due to violence, or a first aider and/or casualty becomes at risk due to violence, you must protect yourself and call for help. Your skills as a first aider are valuable only when the area is safe.

Clean up

After an emergency, it is important to clean-up the area and equipment used properly. Any hard surfaces should be disinfected. Fabrics, where possible, should be laundered. Porous surfaces or materials that cannot be laundered may need to be disposed of.

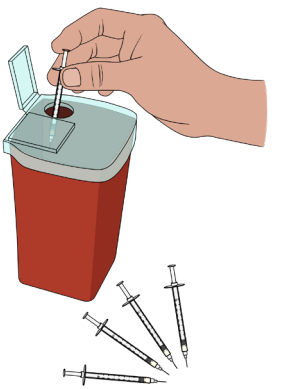
- Items intended to be reused (scissors, forceps) should be wiped of blood and fluids, immersed in a 10% bleach solution (or other disinfecting solution) for 10 minutes, then rinsed and dried
- One-time use items (gauze, gloves) should be put into a garbage bag and tied. That garbage bag can then be put into the regular garbage.
- Any surfaces contaminated by blood or other fluids should be cleaned with a bleach solution or other disinfecting solution

Sharps

In an emergency, sharp objects (or "sharps") may be the cause of the injuries, or used in the first aid. It is important to dispose of these sharp objects properly for both the safety of first aid providers and others. Sharps can include needles, knives, and broken glass. These items may contain contaminated blood and can cut the first aider, exposing them to the contamination.

Cleaning up glass should always be done with mechanical means such as a broom and dustpan. The cleaned up glass should be placed in a puncture-proof container like a cardboard box.

When handling knives, always grasp by the handle and carried blade down. Cleaning knives should follow the steps above for hard surfaces.



Needles should be disposed of in a sharps container. These plastic containers have thick walls and a secured lid that prevents accidental punctures. Ambulances will carry sharps containers and any needles found or used during an emergency can be placed in these containers. Always handle needles by the barrel (the plastic part with the plunger) and never try to re-cap a used needle.

Needles must never be disposed of into the general garbage.

Help at the emergency scene

Bystanders should be asked to leave unless asked to stay and assist.

Other first aiders may offer to help. Identify yourself and accept their assistance. If someone is more qualified to handle the situation, you may ask that person to take control.

First responders include ambulance personnel, police officers and firefighters. They will take charge of the scene as soon as they arrive.

Other authorities may be called to the scene (e.g. an electrical utility crew may arrive if there are downed power lines). Identify yourself and continue giving first aid.

Off-duty doctors, nurses and other health professionals may identify themselves and offer to help.

Ten ways a bystander can help

1. Make the area safe
2. Find all the casualties
3. Find a first aid kit
4. Control the crowd
5. Call for medical help
6. Help give first aid, under your direction
7. Gather and protect the casualty's belongings
8. Take notes
9. Reassure the casualty's relatives
10. Lead the paramedics to the scene of the emergency

Signs and symptoms and mechanism of injury

When referring to injuries, first aiders need to understand **signs and symptoms.**

- A **sign** is something we can see, feel, hear or smell (e.g. bleeding, bruising, agonal breathing, skin discolouration).
- A **symptom** is something the casualty is feeling (e.g. nauseous, weakness, pain) and must tell you.

Mechanism of injury encompasses both what happened to the casualty, and how the injury has affected the casualty. It identifies the cause of the injury. Mechanisms of injury that require an ambulance right away:

- A fall from 6.5 meters (20 feet) or more
- A vehicle collision with signs of a severe impact
- Severe damage to the inside of the vehicle, a bent steering wheel, or a broken windshield
- Casualty was thrown from a vehicle
- The vehicle has rolled over
- Casualty was struck by a vehicle
- Crush injuries

When any of these mechanisms are apparent, call an ambulance as soon as you can. When we understand the cause of the injury, we are able to predict what injuries may be present and what injuries are not likely, even in situations in which there are no visible signs of injury and/or the casualty is unable to describe their symptoms.

Signs and symptoms

Examples of Signs and Symptoms	
Signs you can see	Blood, deformity, bruising, unequal pupils, painful expression and/or flinching, sweating, wounds, unusual chest movement, skin colour, swelling, foreign bodies, vomit, incontinence
Signs you can hear	Noisy or distressed breathing, groans, sucking wounds (chest injury), bones scraping together, quality of speech
Signs you can feel	Dampness, skin temperature, swelling, deformity
Signs you can smell	Casualty's breath (fruity breath, acetone/nail polish breath, or alcohol), vomit, incontinence, gas fumes, burning, solvents or glue
Symptoms the casualty may tell you about	Pain, fear, heat, cold, loss of normal movement, loss of sensation, numbness, tingling sensation, thirst, nausea, faintness, stiffness, feeling faint, weakness, memory loss, dizziness, sensation of a broken bone

The importance of medical help

In first aid, any type of medical care is referred to as medical help. Unless an injury is very minor, you should always make sure the casualty receives medical help following first aid. Medical help may be given at the scene, en route to a medical facility, or in a hospital.

Know the EMS telephone number for your community. This is often 9-1-1 in urban areas. If you are outside of your community, find the EMS phone numbers in the first few pages of the telephone book, or search online.

Calling for medical help is important. The period immediately following a severe, life-threatening injury is known as the **golden hour**. This time is "golden" because the faster a casualty makes it to a hospital emergency room or operating room, the better the chances of survival and recovery.

You can ask a bystander to call for medical help. Provide the person with:

- Necessary phone number
- A description of the casualty's condition
- Directions to follow to reach the scene
- Instructions to report back to you after getting medical help

If you are alone, you must decide whether to stay with the casualty or leave to get help. The correct decision will depend on the specifics of the situation. If you have a mobile phone, call from the scene and perform first aid with the dispatcher's assistance.

Medical care

As a first aider you are not trained to diagnose the nature and extent of an injury or illness; a medical doctor has the training to do this. As a rule, make sure the casualty receives medical care following first aid. For minor injuries, this may not be necessary. **Medical care** is either given by a medical doctor or under the supervision of a medical doctor. Paramedics give medical care because they work under the supervision of medical doctors. Medical care is given in hospitals but it can also be given at the emergency scene or on the way to a medical facility.

1 Call an ambulance or drive the casualty to the hospital?

Always call an ambulance if you can; only transport the casualty to medical help yourself if that is the only possible way to get medical help. Transporting an injured person is often difficult and time-consuming. An ambulance or other rescue vehicle is well-equipped, and the casualty can begin receiving medical help as soon as it arrives.

When an ambulance arrives, do not stop the first aid you are providing until the crew has arrived to the casualty and indicates they are ready to take over. Give a short report to the ambulance crew on the situation; the condition of the casualty; and what you have done so far.

Use **MIST** to help remember what to report:

M – Mechanism of Injury

I – Injuries or illnesses found

S – Signs and symptoms

T – Treatment (first aid) provided so far

The Good Samaritan principles only protect you when giving care at the scene of the emergency, or while transporting the casualty when this is needed to save the casualty's life and medical help is not available. Transporting the casualty unnecessarily leaves you liable if it results in further injury should an accident or incident occur while en route to a hospital or medical station.

Good communication

Communication is necessary in every emergency situation, regardless of the details. As the first aider, there are many people you may need to communicate with – the casualty, bystanders, family members, other first aiders, EMS providers, and other professionals (e.g. police, fire, hydro). Effective communication skills will help you to assess the casualty's condition, and explain what you are doing and why.

Some rules for effective communication:

- Be calm and direct
- Be respectful
- Do not use medical terms
- Call the casualty by name
- Do not diagnose the casualty's condition
- Always be honest, reassuring, and choose your words carefully

As a first aider, the first thing you do when you arrive at an emergency scene is take charge of the situation. You stay in charge until you hand control of the scene over to more qualified people. While in charge, many other people may offer to help.

When handing the scene over to someone other than the casualty, describe the complete history of the incident and pass along any notes you have taken. Be sure to include:

- Your name
- The time you arrived
- The history of the illness or injury, including signs and symptoms observed
- What first aid has been given
- Any changes in the casualty's condition since you took charge

Principles of communication

Though each situation is different, the following general guidelines help improve communication.

1

Focus

Maintain your attention on the casualty. Position yourself at eye level and maintain eye contact.

Terminology

Refrain from using medical terminology when communicating with the casualty or bystanders. Explanations and answers must be clear, concise and easily understood.

Body Language

Refrain from using body language that could be perceived as threatening or aggressive.

Professionalism

Always maintain your professionalism. Explain everything you are doing and why. If what you are doing may be painful, let the casualty know.

Barriers to communication

Despite following the principles of communication above, there are certain barriers that may arise making communication difficult.

- **Language** – the casualty or bystanders may not speak the same language as the first aid provider
- **Physical** – the casualty or bystanders may have a hearing, speech, or visual impairment
- **Cognitive** – the casualty or bystander may not understand the questions or requests
- **Cultural** – different cultures approach interactions with others which may impact communication
- **Environment** – noisy situations can make communication very difficult
- **Technical** – failure or limitations of communication devices (radios, phones) can hinder communication

When faced with these barriers, the first aid provider may have to attempt several different ways to gather information or give directions.

First aid providers should also remember to keep things simple, clear, and to the point. Drawn out descriptions using large words can make it very difficult for the casualty or those around to understand. An example would be "Get me the AED" instead of "I need you to find an Automated External Defibrillator so I can perform cardiopulmonary resuscitation."

Injuries and illnesses

Injuries

When something from outside the body damages tissues, the damaged area is called an injury. How serious an injury is depends on:

- **What tissues are injured**—an injury to a vital organ, or tissues of a vital system, like the nervous system, is serious
- **How bad the injury is**—for instance, a bone broken in half may not be as serious as the same bone shattered into many pieces
- **How much tissue is injured**—a burned hand may be more serious than a burned finger Injuries and energy

Injuries and energy

Injuries result from too much energy being applied to the body. For instance:

- A thermal burn is caused by too much heat energy
- An acid burn is caused by too much chemical energy
- Snow-blindness is caused by too much light energy
- A broken bone is caused by too much mechanical energy
- A stopped heart from an electric shock is caused by too much electrical energy

The body can take a certain amount of energy without being injured. But too much of any sort of energy will cause injury. Three factors determine whether an injury will occur. They are:

- How intense the energy was
- How long the energy was applied to the body
- What part of the body the energy was applied to

1

Most injuries are caused either by something hitting the body or the body hitting something—this is mechanical energy. When something moves, it has mechanical energy. How much mechanical energy something has depends on how fast it is moving and how much it weighs.

Illness

We often think of first aid in the context of injuries only. But when someone becomes very sick, the result can be a medical emergency in which first aid can save a life.

Some illnesses, like heart attacks or strokes come on very fast. Other illnesses progress more slowly and it can be hard to decide exactly when you have a medical problem that calls for a doctor's attention.

Get medical help when any of the following is present:

- Sudden severe pain in any part of the body
- Sudden changes in vision, headache or dizziness
- Severe or persistent diarrhea or vomiting
- Sudden weakness or slurred/jumbled speech
- Persistent high temperature
- Changes in level of consciousness
- Rash of unknown origin
- Repeated fainting
- Obvious depression, suicide threats or attempts
- Whenever you are very worried about yourself or someone in your care

If the casualty is an infant, the following are also reasons to get medical help (in addition to the reasons above):

- The baby has had a seizure
- The baby is blue or very pale
- You think the baby is having trouble breathing
- The baby cries a lot, or won't stop crying

Stress management in emergency situations

First aiders may experience a certain level of stress as a result of the assistance they provide. Stress is the body's normal reaction to physical and psychological events. It can be seen in certain attitudes and behaviours in both casualties and first aiders. It is a biological response and may be reflected in:

- An increase in heart rate
- An increase in blood pressure
- An increase in blood sugar
- Dilation of the bronchi and pupils

Possible reactions of casualties

Casualties may react to stressors in different ways and first aiders must observe and adjust to such reactions which can include:

- **Denial**—the casualty may deny the seriousness of the situation and refuse assistance
- **Resignation**—the casualty may be resigned to dying even if their life does not seem to be in danger, and doesn't want to make any effort to do what is needed
- **Aggressiveness**—the casualty may be hostile
- **Assertiveness**—the casualty is positive, cooperative, and may even want to take charge of their own care including directing the first aiders

Stress management

Managing stress in an emergency situation can make a significant difference in the quality of first aid provided. Appropriate mental preparation and regular first aid skill practice can help first aiders react effectively when faced with an emergency situation. The negative impact of stress can be reduced by understanding it and taking measures to try and overcome it. After serious incidents, it is important for first aiders to process their emotions.

When faced with a highly stressful situation some first aiders may experience prolonged effects of stress and they should seek medical assistance.

Chapter 2



Emergency Scene Management



- Four steps in ESM
- Step one: scene survey
- Step two: primary survey
- Step three: secondary survey
- Step four: ongoing casualty care
- Shock
- Fainting
- Multiple casualty management (triage)
- Lifting and moving

Chapter 2 Emergency scene management

2

Emergency scenes generally begin with confusion as people realize there is an emergency unfolding in front of them. No one knows what to do first, who should be in charge, or how they can help. In this situation, the first aider needs to follow a sequence of actions that ensures safe and appropriate first aid is given and everyone's safety is protected. First aiders use emergency scene management (ESM) to do this. Emergency scene management is the sequence of actions you should follow to ensure safe and appropriate first aid is given.

Steps of ESM

1. **Scene survey**—during the scene survey you take control of the scene, find out what happened and make sure the area is safe before assessing the casualty.
2. **Primary survey**—assess each casualty for life-threatening injuries and illnesses, call or send someone to call 9-1-1, and give life-saving first aid.
3. **Secondary survey**—the secondary survey is a step-by-step way of gathering information to form a complete picture of the casualty's overall condition.
4. **Ongoing casualty care**—during ongoing casualty care you continue to monitor the casualty's condition until medical help takes over.

These steps are generally done in the order above. The initial scene survey, primary survey and the start of life-saving first aid usually happens within one or two minutes. The secondary survey is not always necessary.

Scene Survey

- Take charge of the situation
- Call out for help to attract bystanders
- Assess hazards and make the area safe
- Find out the history of the emergency, how many casualties there are and the mechanism(s) of injury
- Identify yourself as a first aider and offer to help, get consent
- Assess responsiveness

Send or go for medical help as soon as you identify a serious problem and then begin the primary survey. If you have a mobile phone, you can dial 9-1-1 or your local emergency number, and put the device on speaker phone, if possible. This allows the first aider to remain with the casualty.

Primary Survey

Check for life-threatening conditions, the ABCs:

A = Airway

B = Breathing

C = Circulation

The sequential steps of the primary survey should be performed with the casualty in the position found unless it is impossible to do so.

The primary survey should begin immediately after the scene survey.

Check the airway

If the casualty is conscious, ask "what happened?" How well the casualty responds will help you determine if the airway is clear. Use a head-tilt-chin-lift to open the airway of an unresponsive casualty.

If you suspect a head or spinal injury, **and have been trained**, use a jaw-thrust without head-tilt.

Check for breathing

- If the casualty is conscious, check by asking how their breathing is.
- If the casualty is unconscious, check for breathing for at least five seconds, and no more than 10 seconds. If breathing is effective, move on to check circulation. If breathing is absent or ineffective (gasping and irregular, agonal), begin CPR.

Check circulation

- Control obvious, severe bleeding
- Check for shock by checking skin condition and temperature
- Check with a rapid body survey for hidden, severe, external bleeding and signs of internal bleeding

Rapid body survey

The rapid body survey is a quick assessment of the casualty's body which is performed during the primary survey. By running your hands over the casualty's entire body from head to toe (and under heavy outwear), you are able to feel for severe bleeding, internal bleeding and any obvious fractures.

When performing the rapid body survey:

- Wear gloves when possible, and check gloves for blood every few seconds
- Be careful not to cause any further injuries while performing the survey
- Look at the casualty's face to notice any responses to the rapid body survey

Provide first aid for life-threatening injuries or conditions.

- Maintain an open airway with a head-tilt chin-lift or by placing the unresponsive breathing casualty into the recovery position
- Provide CPR if the unresponsive casualty is not breathing or not breathing normally (gasping)
- Control severe bleeding
- Provide support for obvious fractures
- Give first aid for shock by providing first aid for life-threatening injuries and maintaining the casualty's body temperature
- Evaluate the situation and decide whether to do a secondary survey

Do a secondary survey if:

- The casualty has more than one injury
- Medical help will be delayed more than 20 minutes
- Medical help is not coming to the scene and you have to transport the casualty

If you do not do a secondary survey, steady and support any injuries found and give ongoing casualty care until medical help arrives.

How to turn a casualty face up

You should give first aid in the position in which the casualty is found as much as possible. But sometimes you have to turn a casualty over to assess for life-threatening injuries or to give life-saving first aid

1. Extend the arm closest to you over the head.



2. Tuck the far arm against the casualty's side.



3. Cross the far foot over the near foot.



4. Support the head and neck. Firmly grip the clothing at the waist. Roll the casualty over



5. Position the casualty for giving first aid



ESM when a head or spinal injury is suspected

If you suspect a head or spinal injury, protect the head and neck from any movement. Head or neck movement could result in life-long disability or death. Adjust your first aid to this situation as shown below

1. As soon as you see there might be a head or spinal injury, tell the casualty not to move



2. Once you have consent to help the casualty, steady and support the head and neck. Keep elbows firmly supported on thighs or ground. Then, assess responsiveness.



3. If there is a bystander to help, show them how to support the head and neck so you can continue your assessment.



4. Continue your assessment.



5. If a second bystander is available, show them how to steady and support the feet to prevent movement.



6. Keep the head and neck supported (and the feet if possible) while giving further first aid until handover to medical help.



When **moving a casualty** with a suspected head or spinal injury, move them as a unit as much as possible. This means rolling the head, trunk and legs together, or lifting the whole body at the same time. Do what you can to prevent movement.

Turning a casualty face up when a head or spinal injury is suspected

You should give first aid in the position in which the casualty is found as much as possible. But sometimes you have to turn a casualty over to assess for life-threatening injuries or give life-saving first aid

When you suspect a head or spinal injury, turn the casualty as a unit so the head and spine stay in the same relative position

1. The first aider at the head supports the head—placing the right hand along the right side of the casualty's head and the left hand along the left side



2. The other first aider extends the casualty's near arm over their head and gets a good grip on the casualty at the shoulder and waist.
3. At the same time, the two first aiders roll the casualty towards the second first aider.



4. If extra help is available, have the third first aider support the legs to prevent twisting of the neck and spine. With a fourth, put one first aider at the shoulders and another at the waist.



Secondary survey

A **secondary survey** follows the primary survey and any life-saving first aid. It is a step-by-step way of gathering information to form a complete picture of the casualty. In the secondary survey, the first aider is looking for injuries or illnesses that may not have been revealed in the primary survey. You should complete a secondary survey if:

- The casualty has more than one injury
- Medical help will be delayed for 20 minutes or more
- You will transport the casualty to medical help

The secondary survey has four steps:

1. History
2. Vital signs
3. Head-to-toe exam
4. First aid for any injury or illness found

History

A **SAMPLE** history is used to gather a brief medical history of the casualty. This information may be useful for health care professionals who will continue to assist the casualty. If the casualty is unable to respond, some of the SAMPLE history could be answered by a close family member.

S = **symptoms** – what the casualty is feeling (such as pain, nausea, weakness, etc.)

A = **allergies** – any allergies, specifically allergies to medications

M = **medications** – any medications or supplements they normally take, have taken in the past 24 hours, or any doses they may have missed

P = **past or present medical history** – any medical history, especially if it is related to what they are experiencing now. Ask if they have medical alert information

L = **last meal** – last meal they ate and when, anything else taken by mouth

E = **events leading to the incident** – what was happening before the injury/illness? How did the injury occur?

Vital signs

There are four vital signs to check on the casualty

1. Level of consciousness (LOC)
2. Breathing
3. Pulse
4. Skin condition and temperature



Check your applicable workplace legislation/ regulations

Level of consciousness (LOC)

A common method of obtaining a casualty's LOC is using the acronym **AVPU**. When using AVPU to indicate LOC, it is a scale which ranges from good (A), to not as good (V), to bad (P), to worse (U)

A = **Alert** – An alert casualty will have their eyes open and will be able to answer simple questions. An alert casualty is oriented to person, place and time.

V = **Verbal** – The casualty will respond when spoken to, but may not be able to effectively communicate. They may not be oriented to person, place or time.

P = **Pain** – This casualty will only respond when a painful stimuli is delivered, such as pinching them or rubbing your knuckles on their sternum. They may move or make noise, but they will not communicate.



U = **Unresponsive** – the unresponsive casualty will not respond to any stimulus.

Please note that an alternative to quickly estimate a casualty's LOC is to evaluate their eye, verbal and motor skills. If their eyes are open, they can clearly speak, and obey a command such as "squeeze my fingers," they are alert

Breathing

To assess the breathing rate, watch the casualty closely for a total of 30 seconds. It is OK to place your hand on their upper abdomen to feel the rise and fall. Check the quality of the breathing. Carefully count each breath over the 30 seconds and multiply that number by two for breaths per minute.



Normal breathing rates			
Age	**Too slow**	**Normal**	**Too fast**
Infant	Below 25	30-50	Above 60
Child	Below 15	20-30	Above 40
Adult	Below 10	10-20	Above 30

Pulse

The pulse rate is the number of beats your heart takes in one minute, and it is an essential skill for assessing all casualties. The most common places to assess a pulse is at the wrist or neck, and for infants, the inside of the upper arm.

To assess the pulse, use two fingers and gently place them on either the inside of the wrist (just below the hand on the thumb side), or on the side of the neck (carotid artery), or for infants, the inside of the upper arm, on the brachial artery. Press just gently enough to feel the pulse. You may have to feel around the area until you find it. Once you have found the pulse, count the number of beats over 30 seconds and multiply that number by two.



Normal pulse rates	
Age	**Normal pulse range**
Infant	120-150
Child	80-150
Adult	60-100

Skin condition and temperature

When assessing the skin we look for the temperature (warm or cold), the colour (normal skin tones or pale) and whether the skin is dry or wet. Use the back of your gloved hand to feel the casualty's forehead and cheeks. If their skin normal, they will have normal skin colour, and their skin will be warm and dry. If the skin is pale, cold and wet (sweaty), this could be an indication of shock.

Head-to-toe exam

The head-to-toe exam is a complete and detailed check of the casualty for any injuries that may have been missed during the rapid body survey. **Do not examine for unlikely injuries.** You may need to expose an area to check for injuries, but always respect the casualty's modesty and ensure you protect them from the cold. Only expose what you absolutely have to.

- Ask the casualty if they feel any pain before you start. Note any responses.
- Speak to the casualty throughout the process. Explain what you are checking for as you proceed.
- Always watch the casualty's face for any facial expressions that may indicate pain.
- Do not stop the exam. If you find an injury, note it and continue.
- Do not step over the casualty. If you need to, walk around them.
- During a detailed exam, you are looking for all bumps, bruises, scrapes, or anything that is not normal.
- If the casualty is unconscious, look for medic alert devices during your survey, such as a tag, bracelet, necklace, watch, or other indicator.
- Look, then feel



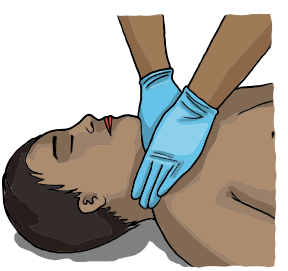
Start at the head:

- Check the skull for anything abnormal
- Check the ears for fluid
- Check the eyes, are the pupils the same size?
- Check the nose for drainage
- Check the mouth, are the teeth intact? Are the lips blue or pale?



Check the neck:

- Are the neck veins bulging?
- Is there a medical alert necklace?
- Check the collarbones
- Check the shoulders on both sides



Check the arms:

- Check each arm completely
- Check the fingernails for circulation by squeezing and watching the blood return

- Ask the casualty to squeeze two of your fingers in both hands at the same time. Do they have an adequate strength and is the strength equal?



Check the chest and under:

- Does it hurt the casualty to breathe?
- Does the chest rise and fall with breaths as it should?
- Reach around the back as far as you can



Check the abdomen and under:

- With flat hands, check the abdomen carefully
- Do not push into the abdomen. Gently feel for pain, tenderness, or rigidity
- Place a flat hand on their abdomen and ask the casualty to push against it. Does this cause pain?
- Reach around the back as far as you can



Check the pelvis:

- Place your hands on top of the pelvic bones and very gently squeeze for stability



Check the legs, ankles, and feet:

- Check each leg completely one at a time
- Is one leg shorter than the other?
- Carefully check the stability of the kneecap and under the knee
- Squeeze or pinch a foot. Ask the casualty what you just did to see if they answer correctly.
- Place both hands on both feet. Ask the casualty to push and then pull against you. Feel for equal strength. Ask the casualty to wiggle their toes and watch for the response.
- Check circulation



First aid for injuries found

When you have completed your exam, give appropriate first aid for any injuries or illnesses found. If the casualty has more than one injury, give first aid to the more serious injuries first.

Document

Upon completion of the secondary survey, document your findings as accurately as possible. This information may be valuable to medical professionals who will continue to assist the casualty.

Documentation is also important in a workplace emergency as it may be used as part of an investigation. Documentation of the incident and the first aid given should be completed on pre-printed forms and maintained as required by provincial regulations/ legislation for reference by investigators.

Ongoing casualty care

Once first aid for injuries and illnesses that are not life threatening has been given:

2

- The first aider will hand over control of the scene to the casualty, or someone else, and end their involvement in the emergency
- The first aider will stay in control of the scene and wait for medical help to take over, or
- The first aider will stay in control of the scene and transport the casualty to medical help

The first aid must maintain the casualty in the best possible condition until handover to medical help by:

- Giving first aid for shock
- Position the casualty based upon their condition
- Monitoring the casualty's condition
- Recording the events of the situation
- Reporting on what happened to whoever takes over

Instruct a bystander to maintain manual support of the head and neck (if head/spinal injuries are suspected). Continue to steady and support manually, if needed.

Recovery Position

This position protects the casualty and also reduces bending and twisting of the spine. This position protects the airway if you must leave the casualty.

1. Position the arm closest to you at 90 degrees in front of the casualty, keeping it out of the way when rolling them.



2. Position the arm furthest from you on the casualty's chest. Bend the far knee.



3. Reach behind the casualty's shoulder and roll casualty towards you by pulling on the far knee.



4. Adjust the position of the arms and leg so the casualty is in a stable position. Place the far arm at 90 degrees to the casualty with the palm down.



5. Give ongoing care.



After the handover

In first aid, we prepare ourselves to care for an injured or ill person. We don't often think about what happens after the casualty has left our care. Immediately following the handover of the casualty you may have a number of practical details to attend to. These details can include cleaning up after the emergency, correcting any unsafe conditions that caused the injury, or making a report on the incident and your involvement.

Once these practical matters are out of the way, we expect things to "return to normal." However, you will likely find yourself thinking about the situation and the details of what happened while you were involved. Following a stressful event, many people review the details and try to evaluate what they did and how they could have done it better.

This reviewing of the events is completely normal and you can expect it to happen. But if thoughts of the incident continue for many weeks, or if they affect your day-to-day life, you may be experiencing the negative effects of critical incident stress (CIS).

Critical incident stress is a common reaction to a stressful emergency situation. The effects of CIS can interfere with your daily life—your job, your relationships, your peace of mind. If this happens to yourself, you need to do something about it, and help is readily available. Start by talking to your family doctor or a doctor at a walk-in clinic. A doctor will understand what you are going through and will suggest a course of action.

The effects of critical incident stress can appear many weeks, months or years after the event.

Shock

Shock is a circulation problem where the body's tissues don't get enough oxygenated blood.

Shock is a danger because any physical injury or illness can be accompanied by shock, and it can quickly progress into a life-threatening condition. Pain, anxiety and fear do not cause shock, but they can make it worse, or make it progress faster. This is why reassuring a casualty and making them comfortable is important.

Medical shock should not be confused with electrical shock or being shocked and surprised. Medical shock is life-threatening, as the brain and other organs cannot function properly.

The following information provides some causes of shock. Severe shock can also result from medical emergencies such as diabetes, epilepsy, infection, poisoning or a drug overdose.

*For casualties with dark skin colour, the colour changes may be observed in the following areas of the body: lips, gums and tongue, nail beds and palms, earlobes, membrane of the inner eyelid.

Common causes of severe shock	
Cause of shock	**How it causes a circulation problem**
Severe bleeding - internal or external (includes major fractures)	Not enough blood to fill blood vessels
Severe burns	Loss of blood plasma (fluid) into tissues—not enough blood to fill blood vessels
Crush injuries	Loss of blood and blood plasma into tissues—not enough blood to fill blood vessels
Heart attack	Heart is not strong enough to pump blood effectively
Spinal cord or nerve injuries	Brain can't control the size of the blood vessels—the blood can't get to the tissues properly
severe allergic reactions	Many things can be affected—breathing, heart function, etc.

Signs and symptoms of shock	
Signs	**Symptoms**
• Pale skin at first, turns blue-grey*	• Restless
• blue-purple lips, tongue, earlobes, fingernails	• Anxious
• Cold and clammy skin	• Disoriented
• Breathing shallow and irregular, fast or gasping for air	• Confused
• Changes in level of consciousness	• Afraid
• Weak, rapid pulse—radial pulse may be absent	• Dizzy
	• Thirsty

First aid for shock

The following actions will minimize shock:

1. Give first aid for the injury or illness that caused the shock
2. Reassure the casualty often
3. Minimize pain by handling the casualty gently
4. Loosen tight clothing at the neck, chest and waist
5. Keep the casualty warm, but do not overheat — use jackets, coats or blankets if you have them
6. Moisten the lips if the casualty complains of thirst. Don't give anything to eat or drink. If medical help is delayed many hours, give small amounts of water or clear fluids to drink — make a note of what was given and when
7. Place the casualty in the best position for their condition Continue ongoing casualty care until handover

The above first aid for shock may prevent shock from getting worse. Whenever possible, add these steps to any first aid you give.

Positioning a casualty in shock

Putting the casualty in the right position can slow the progress of shock and make the casualty more comfortable.

The position you use depends on the casualty's condition. The casualty should be as comfortable as possible in the position you use.

No suspected head/spinal injury; fully conscious

Place the casualty on their back, if injuries permit. Once the casualty is positioned, cover them to preserve body heat, but do not overheat.

No suspected head/spinal injury; less than fully conscious

Place the casualty in the recovery position. When there is decreased level of consciousness, airway and breathing are the priority—the recovery position ensures an open airway.

Conscious with a breathing emergency and/or chest pain

If a conscious casualty is experiencing chest pain or is having difficulty breathing, have them sit in a semi-sitting position, or any position that makes breathing easier for them.

Suspected head/spinal injury

If you suspect a head or spinal injury, steady and support the casualty in the position found. This protects the head and spine from further injury. Monitor the ABCs closely.

As injuries permit

A casualty's injuries may not permit you to put them into the best position. Continue to support the head and neck and, if needed, use a head-tilt chin-lift to maintain the open airway. Always think of the casualty's comfort when choosing a position.

Fainting

Fainting is a temporary loss of consciousness caused by a shortage of oxygenated blood to the brain. Common causes of fainting include:

- Fear or anxiety
- Lack of fresh air
- Severe pain, injury or illness
- The sight of blood
- An underlying medical problem
- Fatigue or hunger
- Long periods of standing or sitting
- Overheating

A person who has fainted is unconscious. Place them in recovery position to protect the airway and prevent possible choking. Place the casualty in a comfortable position as they regain consciousness.

First aid for fainting

- Ensure a supply of fresh air and loosen tight clothing at the neck, chest and waist.
- Make the casualty comfortable as consciousness returns and keep them lying down for 10 to 15 minutes. Continue to monitor breathing and consciousness.

Do not assume a person has "just fainted," unless there is a quick recovery. If the recovery is not quick or complete, stay with the casualty until medical help takes over.

If you have to leave to get medical help or you have to give first aid to other casualties, turn the casualty into the recovery position being as careful as you can if there are any injuries.

Feeling faint or "impending faint"

Sometimes when a person is about to faint, there are warning signs. The person:

- Is pale
- Is sweating
- Feels sick, nauseous, dizzy or unsteady

First aid for an impending faint

1. Place the casualty on their back.
2. Ensure a supply of fresh air—open windows or doors.
3. Loosen tight clothing at the neck, chest and waist.
4. Stay with the casualty until they has fully recovered.

Decreased level of consciousness (LOC)

Consciousness refers to the level of awareness one has of themselves and their surroundings. There are different levels of consciousness ranging from completely conscious to completely unconscious. Many injuries/illnesses can cause changes in a casualty's level of consciousness, including:

- A breathing emergency
- A heart attack
- A head injury
- Poisoning
- Shock
- Alcohol or drug abuse
- Medical condition (epilepsy, diabetes, etc.)

Semi-consciousness and unconsciousness are breathing emergencies for casualties lying on their back, because the tongue may fall to the back of the throat and block the airway. Saliva and other fluids can also pool at the back of the throat and choke the person.

A progressive loss of consciousness means the casualty's condition is getting worse. Always monitor a casualty's level of consciousness and note any changes. A first aider can use the acronym AVPU (alert, verbal, pain, unresponsive) to assess and describe levels of consciousness.

Decreased consciousness is always an urgent situation. The person can quickly become unconscious, and this is a breathing emergency. When you recognize decreased consciousness, get medical help as quickly as possible.

First aid for unconsciousness

1. Start ESM. Perform a scene survey. Call or send for medical help as soon as unresponsiveness is determined.

2. Do a primary survey.



3. Do a secondary survey if necessary.
4. Turn the casualty into the recovery position, if injuries permit. Give ongoing care.

If injuries make it necessary for the casualty to be face up, monitor breathing continuously. If necessary, hold the airway open. Always ensure an open airway.

Loosen tight clothing at the neck, chest and waist, and continue ongoing casualty care until handover. Record any changes in level of consciousness and when they happen.

A decreased level of consciousness also requires urgent medical help.

Multiple casualty management (triage)

The process of making decisions at an emergency scene where multiple people are injured is called **triage**. In triage, first aiders quickly examine all casualties and place them in order of greatest need for first aid and for transportation. The idea is to do the most good for the greatest number of casualties.

Casualties are categorized into three levels of priority:

- **Highest priority**—casualties who need immediate first aid and transportation to medical help
- **Medium priority**—casualties who probably can wait one hour for medical help without risk to their lives
- **Lowest priority**—casualties who can wait and receive first aid and transportation last, or casualties who are obviously dead



Note: in the event of a lightning strike, where more than one person is injured, the principles of multiple casualty management are reversed. Give first aid to unresponsive non-breathing casualties since the casualties that are still breathing are recovering.

The first aid priorities for injuries		
Priority	**Condition**	**Causes**
High priority Airway	Foreign body blocking airway	Choking on food
	Tongue or fluids blocking airway	Unconscious, lying on back
	Swollen airway	Allergic reaction, airway infection
High priority Breathing	Injured chest and/or lungs	Chest injury, broken ribs
	Brain not controlling breathing properly	Poisoning, drug overdose, stroke, electric shock
	Not enough oxygen reaching blood	Not enough oxygen in air, carbon monoxide poisoning
High priority Circulation	Severe bleeding	External bleeding or internal bleeding
	Severe shock	Bleeding, serious illness, poisoning
Medium priority Injuries that have potential for life-long disability	Fractures that could affect breathing	Broken ribs, shoulder blade
	Fractures—open, severe or multiple bones	Broken upper leg, pelvis, crushed arm
	Head/spinal injuries	Fall from a six-foot ladder
	Critical burns	Critical burns to the hands
Low priority Minor injuries or obviously dead	Minor fractures	Broken lower leg, lower arm, hand, finger, etc.
	Minor bleeding	Bleeding not spurting or free-flowing
	Non-critical burns	Moderate degree burns to the forearms
	behavioural problems	Grief or panic
	Obviously dead	Obvious massive injuries, no pulse or other signs of circulation

Triage sequence of actions

Begin ESM

- Determine how many casualties there are in the scene survey.

Start with the nearest casualty, and move outward

- Do a primary survey
- Give first aid for life-threatening injuries
- If the person is obviously dead, go to the next nearest casualty

Repeat step 2 for each casualty

- Always move to the next nearest casualty

Categorize

- Decide which casualties have the highest priority, second priority, and lowest priority.

Arrange transportation

- Arrange for the highest priority casualties to be transported to medical help as soon as possible

Perform secondary survey

- Begin with the highest priority. Give appropriate first aid, and move on

Give ongoing care for each casualty until transported

In a multiple casualty situation, constantly assess the casualties and the situation and make changes to priorities.

Lifting and moving

Always try to give necessary first aid where the casualty is found, then wait for the paramedics to move the person. However, there are times when this is not possible.

You may have to move a casualty when:

- There are life-threatening hazards to yourself or the casualty e.g. danger from fire, explosion, gas or water
- Essential first aid for wounds or other conditions cannot be given in the casualty's present position or location
- The casualty must be transported to a medical facility

If life-threatening hazards make it necessary to move a casualty right away, you may need to use a rescue carry.

In urgent and dangerous situations where casualties are moved with less than ideal support for injuries, the casualty's injuries may be made worse by improper movement and handling. The chance of further injury can be reduced with proper rescue carry techniques.

Always move the casualty the shortest possible distance to safety and to provide essential first aid. Use bystanders to help you and support any injuries the best you can during the move. Keep the risks to the casualty, yourself and others to a minimum.

Rolling Cots (Stretchers)

If your workplace uses rolling cots to transport injured workers, it is crucial that you have proper training on how to operate and handle the cot before you use it. Failure can result in the cot tipping or dropping, and causing further injury.

Occupational Hazards

When working in and around occupational hazards such as confined spaces, trenches, machinery, and hazardous gases, workers should know the proper response protocols. Ensure you have the appropriate training for the type or rescue you will be undertaking as well as the proper equipment to keep yourself safe while rescuing another worker.

Helicopters

In some locations, a helicopter may be sent to transport an injured worker to hospital. Anyone working around helicopters should have proper training and everyone should follow these guidelines:

- Never approach without permission of the pilot or crew chief
- Always follow directions from the pilot or crew chief on from which direction to approach the aircraft
- Know the restricted and danger areas around the aircraft and remain outside these areas unless you need to enter
- The tail and tail-rotor of the helicopter poses a significant danger and should be avoided

Transporting a casualty

Generally speaking, first aid providers will not transport casualties to the hospital, leaving that to ambulances. However, there are instances where you may choose to transport someone to the hospital yourself.

- The injuries are minor and the casualty is stable
- The response time for EMS to arrive is prohibitively long (i.e. remote areas)
- If the above conditions are met, and the casualty is a family member or close friend—do not transport strangers to the hospital in your own vehicle whenever possible

Care during transport

Every effort must be taken to ensure that injuries are not made worse while transporting a casualty to a medical facility. Take steps to keep the casualty comfortable and in a position that will not cause unneeded movement of injured limbs. Have another person accompany you to monitor the casualty if possible.

Do not drive fast. If a casualty needs to be transported quickly, an ambulance should be called. Follow all local traffic laws. Find the smoothest route possible to make the ride as comfortable as possible.

Meeting an ambulance crew enroute

Some work locations are isolated and response times of EMS are extended. In these instances, where provincial regulations permit, casualties can be transported toward medical help with the plan to meet an ambulance part way. When selecting a meeting point, be sure the location is clear to both sides and it is easily found. Identify any landmarks or businesses that can aid in finding the location. In the event you arrive before the ambulance, contact EMS and get an update on the ambulance location before moving the meeting location.

Lifting techniques and proper body mechanics

Moving any casualty from an emergency scene poses dangers to the rescuer as well as the casualty. If the casualty must be moved, select the method that will pose the least risk to the casualty and to you. You can be of little help to a casualty if you injure yourself in the rescue.

Using incorrect body mechanics in lifting or moving a casualty may leave the rescuer suffering muscle strains. Use the following lifting guidelines:

- Stand close to the object to be lifted.
- Bend your knees, not your waist.
- Tilt the object so that you can put one hand under the edge or corner closest to you.
- Place your other hand under the opposite side or corner, getting a good grip on the object.
- Use your leg muscles to lift, and keep your back straight.
- When turning, turn your feet first; don't twist your body.

When lowering the object, reverse the procedure.